

Numerical Study of Hydrogen-Enriched Methane–Air Combustion Using Detailed Chemical Kinetics

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Computer Methods in Combustion (MKWS)

June 16, 2026

Abstract

This work investigates the effect of hydrogen enrichment on the combustion of methane–air mixtures using detailed chemical kinetics in Cantera with the GRI-Mech 3.0 mechanism. Hydrogen fraction in the fuel was varied from 0 % to 40 % (molar) across a range of equivalence ratios. Adiabatic equilibrium states, autoignition delay times, laminar flame speeds, and a perfectly stirred combustor were analysed. At $\phi = 1.0$, 300 K and 1 atm, the calculated laminar flame speed increased from 37.68 cm s^{-1} for methane–air to 52.16 cm s^{-1} for 40 % hydrogen in the fuel. At $T_0 = 1000 \text{ K}$, $p = 10 \text{ atm}$ and $\phi = 1.0$, the constant-pressure ignition delay decreased from 81.99 ms to 12.64 ms over the same enrichment range. The perfectly stirred reactor results identify stability-transition brackets between cold extinguished and hot stable steady-state solutions on the tested residence-time grid.

1 Introduction

Hydrogen is widely studied as a clean fuel for gas-fired combustors and turbines, and blending hydrogen into methane is a practical near-term route to lowering carbon emissions from existing equipment [1]. Published experimental and review studies report that hydrogen enrichment generally increases laminar burning velocity and changes ignition and flame stability characteristics. Its implications for flashback and nitrogen-oxide formation depend strongly on mixture composition and operating conditions [1, 2].

The objective of this study is to quantify, with a single consistent kinetic model, how the molar hydrogen fraction in a CH_4/H_2 fuel blend affects: (i) the adiabatic flame temperature and equilibrium product composition; (ii) the autoignition delay time and its sensitivity to temperature, pressure and equivalence ratio; (iii) the laminar flame speed; and (iv) the behaviour of an ideal perfectly stirred combustor on a prescribed residence-time grid.

2 Methodology

2.1 Chemical mechanism

All simulations were performed with Cantera [3] using the GRI-Mech 3.0 mechanism [4], an optimised detailed mechanism for natural-gas combustion comprising 53 species and 325 reactions and including nitrogen chemistry for NO formation. The mechanism is well suited to methane-dominated mixtures; its applicability to high hydrogen fractions is discussed in Section 4.

2.2 Fuel mixture definition

The fuel is a binary CH_4/H_2 blend parameterised by the molar hydrogen fraction

$$x_{\text{H}_2} = \frac{n_{\text{H}_2}}{n_{\text{H}_2} + n_{\text{CH}_4}}, \quad (1)$$

so that the fuel composition is $\{\text{CH}_4 : 1 - x_{\text{H}_2}, \text{H}_2 : x_{\text{H}_2}\}$. Air is modelled as $\text{O}_2 + 3.76 \text{N}_2$, and the mixture is set with a prescribed equivalence ratio ϕ .

2.3 Adiabatic equilibrium

Equilibrium states are obtained at constant enthalpy and pressure (HP problem), giving the adiabatic flame temperature and the equilibrium mole fractions of NO, CO, CO_2 and H_2O . The initial state is $T_0 = 900 \text{ K}$ and $p = 10 \text{ atm}$. The tested equivalence-ratio grid is $\phi = 0.6, 0.8, 1.0, 1.2$ and 1.4 , and the hydrogen fractions in the fuel blend are 0, 10, 20, 30 and 40 mol %.

2.4 Autoignition delay

Autoignition is computed in a homogeneous adiabatic reactor, modelled both as a constant-volume (**IdealGasReactor**) and a constant-pressure (**IdealGasConstPressureReactor**) system. The ignition delay τ is defined as the time of maximum temperature-rise rate, $\max(dT/dt)$; the time of peak OH mass fraction is used as an independent cross-check. The primary temperature sweep uses the constant-pressure reactor at $\phi = 1.0$, $p = 10 \text{ atm}$, $T_0 = 900, 950, 1000, 1050$ and 1100 K , with hydrogen fractions of 0, 10, 20, 30 and 40 mol %. The pressure sweep uses $T_0 = 1000 \text{ K}$, $\phi = 1.0$, pressures of 5, 10 and 20 atm, and hydrogen fractions of 0, 20 and 40 mol %. The equivalence-ratio sweep uses $T_0 = 1000 \text{ K}$, $p = 10 \text{ atm}$, $\phi = 0.6$ to 1.4 in increments of 0.1, and hydrogen fractions of 0, 20 and 40 mol %.

2.5 Laminar flame speed

The laminar flame speed is obtained from a one-dimensional freely propagating premixed flame (**FreeFlame**) with adaptive grid refinement. The final calculations use multicomponent transport with Soret diffusion enabled. The reported speed is the unburned-gas inlet velocity of the converged solution. Reference conditions are an unburned temperature of 300 K and a pressure of 1 atm, chosen so that the stoichiometric methane–air result can be used as a reference consistency check (Section 3.4). The domain width is 0.03 m; the tested equivalence ratios are $\phi = 0.6, 0.8, 1.0, 1.2$ and 1.4 ; and the hydrogen fractions are 0, 20 and 40 mol %. Grid refinement uses ratio = 3.0, slope = 0.06 and curve = 0.12.

2.6 Perfectly stirred reactor

A perfectly stirred reactor (PSR) is built from an inlet reservoir, an **IdealGasReactor** and an exhaust reservoir, coupled by a **MassFlowController** and a **PressureController**. The inlet mass flow is set as $\dot{m} = m_{\text{reactor}}/\tau_{\text{res}}$, allowing the residence time τ_{res} to be swept. The PSR is zero-dimensional, adiabatic and perfectly stirred. It is initialised from equilibrium products at the inlet enthalpy and pressure, then continued from long to short residence time; as a result, the calculated sequence follows the hot burning branch where that branch is available. The steady state is solved at each tested residence time to identify a stability-transition bracket between cold extinguished and hot stable steady-state solutions on the discrete grid. The inlet temperature is 1000 K, the pressure is 10 atm, the tested equivalence ratios are $\phi = 0.7$ and 1.0 , and the hydrogen fractions are 0, 20 and 40 mol %. The residence-time grid is 0.001 to 50 ms as configured, and the hot-gas residence time is $\tau = m_{\text{ss}}/\dot{m}$. Equilibrium-product initialisation is used before continuation. A 500 K temperature rise is used as the operational classification threshold between extinguished and stable cases.

3 Results and discussion

3.1 Adiabatic flame temperature and equilibrium composition

On the tested five-point equivalence-ratio grid, the maximum adiabatic flame temperature is observed at $\phi = 1.0$ for every fuel blend (Figure 1). For methane–air, T_{ad} increases from 2134.18 K at $\phi = 0.6$ to 2595.74 K at $\phi = 1.0$, then decreases to 2427.90 K at $\phi = 1.4$. With 40 mol % hydrogen in the fuel, the corresponding values are 2161.47 K, 2622.25 K and 2480.57 K. Hydrogen enrichment therefore raises the stoichiometric equilibrium temperature by 26.51 K over the tested range.

The equilibrium NO mole fraction is largest near the lean side of the tested range: for methane–air it is 0.007842 at $\phi = 0.8$, 0.004817 at $\phi = 1.0$ and 0.000163 at $\phi = 1.4$. At 40 mol % hydrogen, the corresponding values are 0.008123, 0.004987 and 0.000224. Equilibrium CO is very small on the lean side and increases strongly for rich mixtures, from 0.000256 at $\phi = 0.6$ to 0.078503 at $\phi = 1.4$ for methane–air, and from 0.000266 to 0.067442 for the 40 mol % hydrogen blend. These equilibrium mole fractions must not be interpreted as actual engine emissions: they represent infinite-time adiabatic chemical equilibrium at fixed enthalpy and pressure, without finite residence time, turbulent mixing, heat loss, or exhaust quenching.

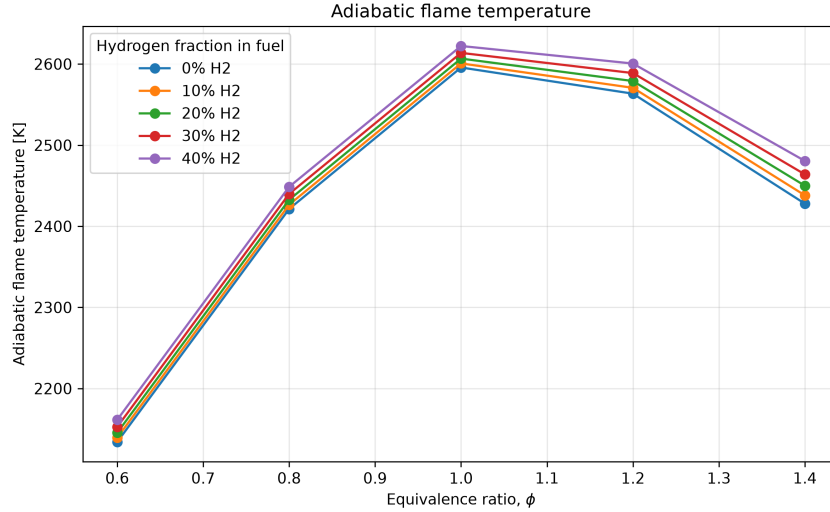


Figure 1: Adiabatic flame temperature versus equivalence ratio for different hydrogen fractions at $T_0 = 900$ K and $p = 10$ atm.

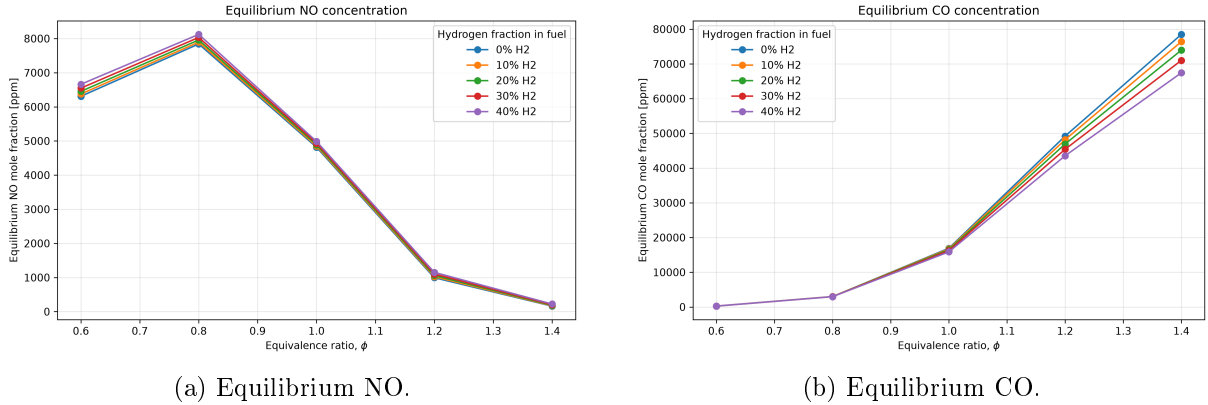


Figure 2: Equilibrium pollutant mole fractions versus equivalence ratio at $T_0 = 900$ K and $p = 10$ atm.

3.2 Autoignition delay

The ignition-delay results show a strong temperature dependence (Figure 3). At $\phi = 1.0$ and $p = 10$ atm, methane–air decreases from 601.98 ms at $T_0 = 900$ K to 17.34 ms at $T_0 = 1100$ K. The 40 mol % hydrogen blend decreases over the same temperature range from 172.58 ms to 1.39 ms. At $T_0 = 1000$ K, hydrogen enrichment reduces the delay from 81.99 ms at 0 mol % hydrogen to 18.09 ms at 10 mol %, 14.34 ms at 20 mol %, 13.21 ms at 30 mol % and 12.64 ms at 40 mol %.

At $T_0 = 1000$ K and $\phi = 1.0$, increasing pressure from 5 atm to 20 atm shortens the methane–air ignition delay from 178.55 ms to 39.61 ms. For 20 mol % hydrogen the same pressure increase reduces the delay from 18.85 ms to 9.98 ms, and for 40 mol % hydrogen from 14.62 ms to 9.24 ms.

The equivalence-ratio sweep at $T_0 = 1000$ K and $p = 10$ atm shows different behaviour across fuel blends (Figure 4). Methane–air lengthens monotonically across the tested range, from 71.76 ms at $\phi = 0.6$ to 93.52 ms at $\phi = 1.4$. The hydrogen-enriched cases shorten as the mixture is enriched over the same range: the 20 mol % case changes from 18.41 ms to 12.37 ms, and the 40 mol % case from 16.36 ms to 10.76 ms. These trends are reported as outcomes of the present finite-rate simulations; detailed radical pathway attribution would require additional reaction-path or sensitivity analysis.

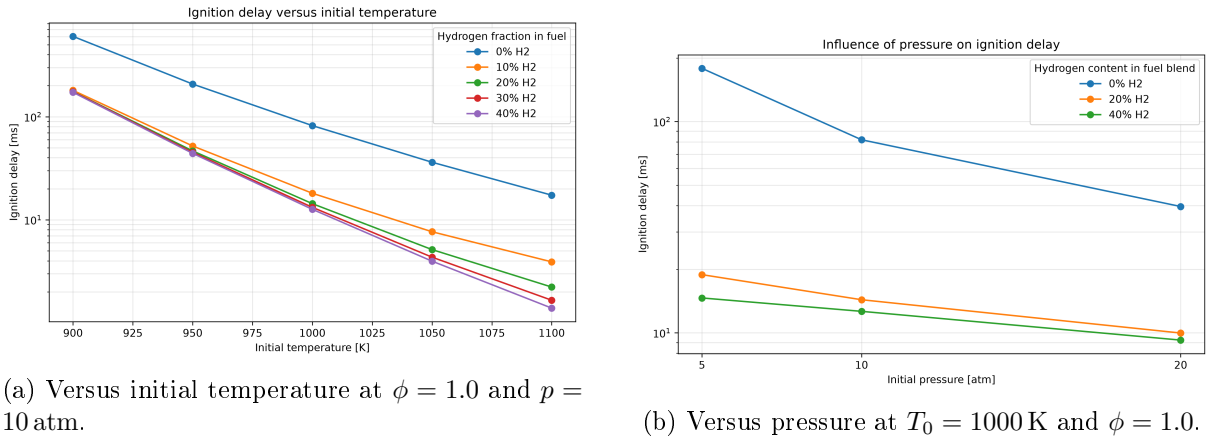


Figure 3: Autoignition delay time for the constant-pressure reactor.

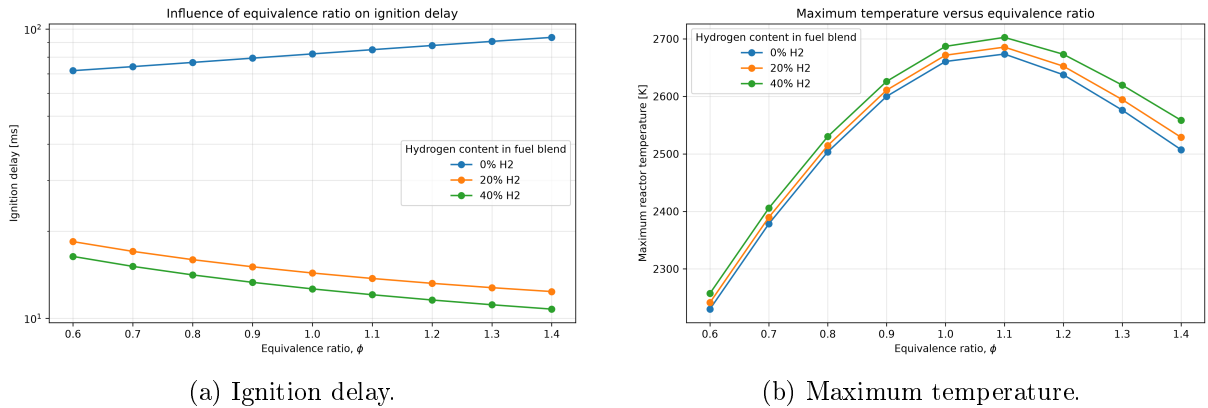


Figure 4: Equivalence-ratio sweep at $T_0 = 1000$ K, $p = 10$ atm and hydrogen fractions of 0, 20 and 40 mol %.

3.3 Constant-volume versus constant-pressure reactors

The reactor-model comparison isolates the effect of the constant-volume versus constant-pressure idealisation for the same mixtures and initial temperatures. For all tested cases, the constant-pressure ignition delay is longer by approximately 2.6% to 8.8%. The smallest difference is 2.60% at $T_0 = 900$ K and 0 mol % hydrogen (586.74 ms constant-volume versus 601.98 ms constant-pressure), while the largest is 8.84% at $T_0 = 1100$ K and 20 mol % hydrogen (2.05 ms versus 2.23 ms). At $T_0 = 1000$ K, the constant-pressure delays are 81.99 ms, 14.34 ms and 12.64 ms for 0, 20 and 40 mol % hydrogen, compared with constant-volume values of 79.74 ms, 13.41 ms and 11.86 ms. The constant-pressure reactor is therefore more representative of the adopted combustor-oriented idealisation, but it is not a complete gas-turbine model.

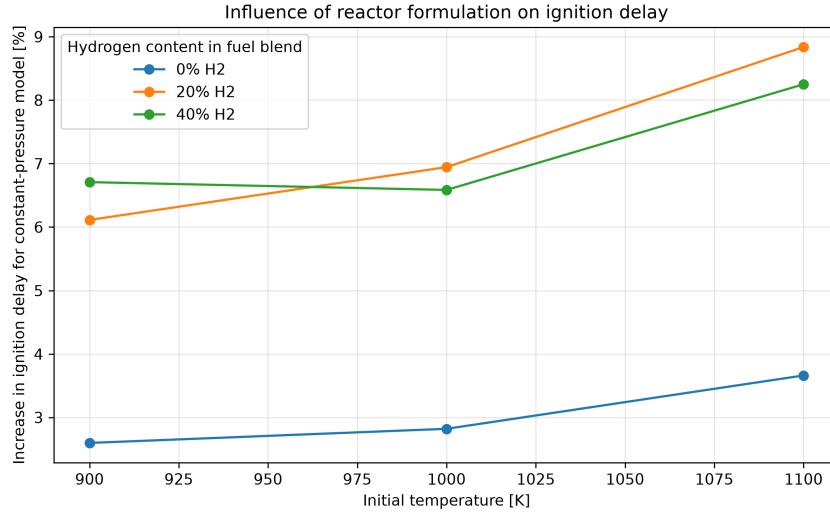


Figure 5: Ignition delay at $\phi = 1.0$ and $p = 10$ atm: constant-volume versus constant-pressure reactor for $T_0 = 900, 1000$ and 1100 K, and hydrogen fractions of 0, 20 and 40 mol %.

3.4 Laminar flame speed

The computed laminar flame speed of stoichiometric methane–air at reference conditions is 37.68 cm s^{-1} . This value is reported as a result of the present **FreeFlame** model; no dedicated experimental validation was performed. Hydrogen enrichment increases the calculated stoichiometric flame speed to 52.16 cm s^{-1} at 40 mol % hydrogen in the fuel (Figure 6).

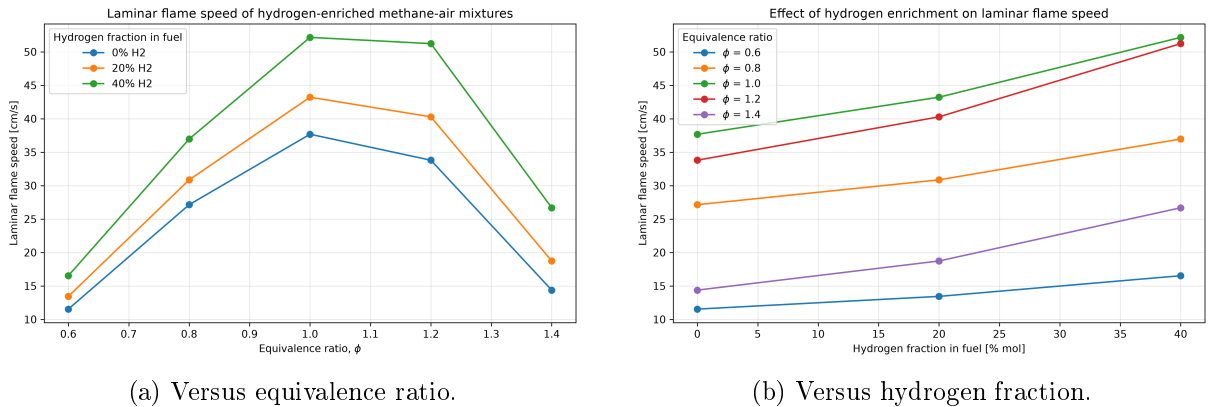


Figure 6: Laminar flame speed of hydrogen-enriched methane–air flames.

3.5 Perfectly stirred reactor stability transition

For each tested fuel blend and equivalence ratio, the PSR calculations provide a stability-transition bracket between cold extinguished and hot stable steady-state solutions on the tested residence-time grid. This ideal PSR calculation does not represent actual engine behaviour, CFD, turbulence, combustor geometry, recirculation or wall heat loss.

Table 1: Discrete PSR stability-transition brackets on the tested residence-time grid.

ϕ	H ₂ in fuel [mol %]	Extinguished τ [ms]	Stable τ [ms]
0.7	0	0.005	0.010
0.7	20	0.002	0.005
0.7	40	0.002	0.005
1.0	0	0.002	0.005
1.0	20	0.002	0.005
1.0	40	0.002	0.005

These values are discrete brackets from the configured residence-time grid, not exact mathematical transition points. Hydrogen enrichment moves the observed bracket for the lean $\phi = 0.7$ case from 0.005–0.010 ms for methane–air to 0.002–0.005 ms for the enriched cases; at $\phi = 1.0$, all three tested fuel blends share the same 0.002–0.005 ms bracket.

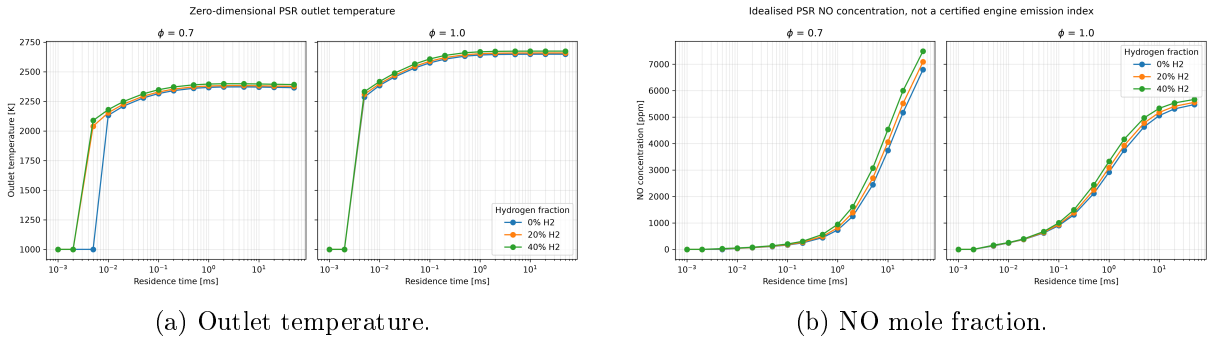


Figure 7: PSR behaviour versus residence time; the drop at short residence time marks a stability-transition bracket on the tested grid. Inlet conditions are $T_{\text{in}} = 1000$ K, $p = 10$ atm, with $\phi = 0.7$ and 1.0.

3.6 Nitrogen oxide formation

The equilibrium and PSR results report NO mole fractions from the GRI-Mech 3.0 chemistry. Established nitrogen-chemistry pathways in combustion are reviewed by Miller and Bowman [5]. The present study does not perform reaction-path, rate-of-production or sensitivity analysis, so it reports only the overall NO trends predicted by the selected mechanism. These values are not certified engine-emission indices.

4 Model limitations

GRI-Mech 3.0 was optimised and validated for natural-gas (methane-dominated) combustion [4]. For the hydrogen fractions considered here ($x_{\text{H}_2} \leq 0.4$), the mixture remains methane-dominated, so its use is defensible; however, for higher hydrogen content a dedicated hydrogen mechanism, such as that of Burke et al. [6] or Conaire et al. [7], would be more appropriate. The flame-speed calculations use multicomponent transport with Soret diffusion, but they still

represent one-dimensional, freely propagating, adiabatic premixed flames. The PSR is zero-dimensional, adiabatic and perfectly stirred, with no representation of actual engine operation, CFD, turbulence, geometry, recirculation or wall heat loss. The equilibrium calculations also assume infinite-time chemical equilibrium and therefore cannot represent finite-rate effects in the nozzle/exhaust.

5 Conclusions

1. At $\phi = 1.0$, 300 K and 1 atm, increasing the fuel hydrogen fraction from 0 % to 40 % increased the calculated laminar flame speed from 37.68 cm s^{-1} to 52.16 cm s^{-1} .
2. At $T_0 = 1000 \text{ K}$, $p = 10 \text{ atm}$ and $\phi = 1.0$, the constant-pressure ignition delay decreased from 81.99 ms for methane–air to 12.64 ms for the 40 % hydrogen blend.
3. For the tested PSR grid, cold extinguished and hot stable steady-state solutions were separated by a stability-transition bracket in residence time, rather than by a resolved exact transition point.
4. The present results are limited to detailed-chemistry idealisations: equilibrium reactors, homogeneous ignition calculations, one-dimensional freely propagating flames and a zero-dimensional adiabatic PSR.

References

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